D1 Problem and research question proposal

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Searching for jobs and hiring are very hard, annoying, and lengthy processes, and everyone has to do them. Companies need employees and people need jobs. Why not make it fun and easy? Tinder is something that almost everyone has used or knows about. A very easy way to find a mate. Can that be applied to other aspects of life? For example, job applications.

We chose this domain because it applies to all of us specifically, as we all are having to apply to internships and jobs. We want to know if we can make the job search more enjoyable for both parties? In this project we hope to learn more about the tinder UI to see if it can be applied to more than just job applications. It connects to a useful analytical problem because people are always trying to make things better and easier, and the tinder UI is a very simple way to implement this.

From this idea, we want to research if a Tinder UI will really make an impact on the job application and hiring process and if it can be applicable to other industries. Does a swipe-card interface lead users to engage with more postings than a traditional list-style interface, without lowering how many of their “interested” choices they still stand by afterward? And does the same advantage hold when the cards are of other industries instead of jobs?

We will use two databases to train our model. The model will learn off of the resume database, and predict off of the job posting database. The last source is a UI using the tinder swiping feature. The job posting dataset is a public kaggle dataset with records of 124,000+ job postings listed in 2023 and 2024. Each individual posting contains dozens of valuable attributes for both postings and companies, including the title, job description, salary, location, application URL, and work-types (remote, contract, etc). The resume dataset is also a public one from

Kaggle, containing 2400+ resumes in PDF format, spanning different categories of jobs. They all contain the person's skills, work experience, and education. The final source is a GitHub housing a Tinder swiping UI for our project to use.

Overall, we are making a recommendation system for job and candidate searching. The application will match job seekers to available jobs via resume and job description, as well as employers to candidates. This will be done via sentence embedding and cosine similarity. Users will be asked to complete a survey for additional features. These questions might ask for salary, location, experience, etc. The model will most likely be a recommendation model that will recommend based on "Good Fit", "Partial Fit", and "No Fit". The best matches will be displayed to the user first. There should also be a tab for mutual likes and a link to apply.

One main risk could be mindless, rapid swiping. To keep the data clean for employers, we can track interaction speed and filter out candidates who are blind-swiping faster than someone can read the posting. The application will likely not have many real users, so training on actual users will need to be clearly labeled and factored in.

After training, the model could be biased toward certain names, locations, writing styles, etc. Any identifiable information needs to be removed or ignored during model training, and make sure matches are evenly distributed across different groups. The main model is meant to best match candidates with jobs and jobs with candidates. This is not a measure of hireability, just if both users are satisfied with their job search.

Sources of datasets + tinder swipe UI:

https://www.kaggle.com/datasets/arshkon/linkedin-job-postings

https://www.kaggle.com/datasets/snehaanbhawal/resume-dataset

https://github.com/3DJakob/react-tinder-card